Smart CCTV with Help Detection and Alarm

A. Introduction

Schools are supposed to be institutions where learning is done safely without threats. However, emergencies including bullying, accidents, health problems, and unauthorized intrusion can occur at any given time. The existing Closed-Circuit Television (CCTV) system records videos which have to be constantly monitored by someone. This may result in a delayed reaction during an emergency situation.

The suggested Smart CCTV with Help Detection and Alarm makes use of artificial intelligence and computer vision to identify a person who requires help and abnormal activity in real time. The system sends an alarm to the security of the school instantly.

B. Problem and Objectives

Problem:

- CCTV is conventional and simply captures footage but doesn't automatically detect emergencies.
- The security team cannot watch everything from all cameras simultaneously.
- A delayed reaction to emergencies might endanger students and teachers.

Objectives:

- Create an intelligent CCTV system that will automatically detect help signals or emergencies.
- Notify security personnel of these in real-time.
- Increase the safety and security of the students, teachers, and security staff.
- Offer support to schools through technology for efficient emergency management.

C. SDGs

SDG 4 – Quality Education

When learning takes place in a safe environment, the learners can be assured of being able to concentrate on their studies. Through the implementation of the above technology system, a safe learning environment will be created.

SDG 16 – Peace, Justice and Strong Institutions

SDG 16 advocates for peaceful and safe societies. The project is meant to enhance the safety of the schools by preventing any form of violence and by enhancing the emergency response. Through the use of intelligent surveillance technology, the schools are able to react promptly to any situation and uphold peace and harmony within the institution.

D. Current Technology

Most schools use standard CCTV cameras that record footage for later review. Some modern systems already include:

- Motion detection
- Face recognition
- Intrusion detection
- Mobile notifications

However, many existing systems cannot recognize when someone is asking for help or experiencing an emergency. Human monitoring is still required most of the time.

E. Proposed Solution

Create a Smart Surveillance Camera with Help Detection and Alarm that detects help gestures or suspicious activities by employing artificial intelligence (AI) image processing techniques.

The Smart Surveillance Camera will:

- Constantly monitor CCTV videos.
- Detect emergency gestures or suspicious activities by AI.
- Alert immediately.
- Notify security guards of the school via a mobile application or computer.
- Store CCTV footage for further investigation.

F. Proposed Process

1. The CCTV cameras keep a close watch on all the school premises.
2. The AI technology evaluates the video stream.
3. In case a help signal or any emergency activity is identified by the system, verification takes place.
4. An alarm gets set off.
5. Security staff get notified instantly.
6. Security team rushes to the spot immediately.
7. The event is documented for reference.

Conclusion

The Smart CCTV with Help Detection and Alarm provides school security by integrating intelligent CCTV cameras with emergency detection in real time. This helps in achieving SDG 4 by making the environment safe for learning while SDG 16 is met by ensuring peace and security.